

REFORMULATION OF THE LEVERAGE FOR LAD REGRESSION

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ABSTRACT. This technical note presents a computational reformulation of the leverage measure proposed by Flores (2015). For each observation i , we have,

$$\gamma_i(\mathbf{X}) = \max_{\|\mathbf{g}\|_2=1} \frac{|\mathbf{x}_i^\top \mathbf{g}|}{\sum_{j=1}^n |\mathbf{x}_j^\top \mathbf{g}|}.$$

We verify that, by homogeneity, this problem is equivalent to a linear programming problem. The absolute values are linearized using auxiliary variables, and the free vector $\mathbf{g}$ is transformed into the difference of two non-negative vectors. Finally, we develop an implementation based on the primal simplex method and *warm start*, taking advantage of the fact that all problems associated with the different observations have the same feasible region.

Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ be the model matrix, whose rows are the p -dimensional vectors $\mathbf{x}_1, \dots, \mathbf{x}_n$. Flores (2015) define

$$\gamma_i(\mathbf{X}) = \max_{\substack{\mathbf{g} \in \mathbb{R}^p \\ \|\mathbf{g}\|_2=1}} \frac{|\mathbf{x}_i^\top \mathbf{g}|}{\sum_{j=1}^n |\mathbf{x}_j^\top \mathbf{g}|}, \quad i = 1, \dots, n. \quad (1)$$

These values quantify the individual leverage of the observations and correspond to a measure that depends solely on the design matrix and not on the response.

The ratio in (1) is homogeneous of degree zero. Thus, for any $a \neq 0$,

$$\frac{|\mathbf{x}_i^\top (a\mathbf{g})|}{\sum_{j=1}^n |\mathbf{x}_j^\top (a\mathbf{g})|} = \frac{|\mathbf{x}_i^\top \mathbf{g}|}{\sum_{j=1}^n |\mathbf{x}_j^\top \mathbf{g}|}.$$

Therefore, the constraint $\|\mathbf{g}\|_2 = 1$ can be replaced by a L_1 normalization of $\mathbf{X}\mathbf{g}$, that is,

$$\gamma_i = \max_{\mathbf{g}} \{|\mathbf{x}_i^\top \mathbf{g}| : \|\mathbf{X}\mathbf{g}\|_1 \leq 1\}.$$

Because the feasible region is symmetric about the origin, we have,

$$\max_{\|\mathbf{X}\mathbf{g}\|_1 \leq 1} |\mathbf{x}_i^\top \mathbf{g}| = \max_{\|\mathbf{X}\mathbf{g}\|_1 \leq 1} \mathbf{x}_i^\top \mathbf{g},$$

this leads

$$\gamma_i = \max_{\mathbf{g}} \{\mathbf{x}_i^\top \mathbf{g} : \|\mathbf{X}\mathbf{g}\|_1 \leq 1\}.$$

To linearize $\|\mathbf{X}\mathbf{g}\|_1$, we introduce $\mathbf{t} \in \mathbb{R}^n$, $\mathbf{t} \geq \mathbf{0}$, such that $\mathbf{t} \geq \mathbf{X}\mathbf{g}$, and $\mathbf{t} \geq -\mathbf{X}\mathbf{g}$, or equivalently

$$\mathbf{X}\mathbf{g} - \mathbf{t} \leq \mathbf{0}, \quad \text{and} \quad -\mathbf{X}\mathbf{g} - \mathbf{t} \leq \mathbf{0}.$$

Then $t_j \geq |\mathbf{x}_j^\top \mathbf{g}|$, and the condition

$$\mathbf{1}^\top \mathbf{t} \leq 1$$

guarantees that $\|\mathbf{X}\mathbf{g}\|_1 \leq 1$.

This formulation allows us to use a standard primal simplex; consider

$$\mathbf{g} = \mathbf{g}_+ - \mathbf{g}_-, \quad \mathbf{g}_+ \geq \mathbf{0}, \quad \mathbf{g}_- \geq \mathbf{0}.$$

and similarly,

$$\mathbf{X}\mathbf{g} = \mathbf{X}\mathbf{g}_+ - \mathbf{X}\mathbf{g}_-.$$

Therefore, the problem can be written as

$$\begin{aligned} & \max_{g^+, g^-, t} \quad \mathbf{x}_i^\top \mathbf{g}_+ - \mathbf{x}_i^\top \mathbf{g}_-, \\ & \text{subject to:} \quad \mathbf{X}\mathbf{g}_+ - \mathbf{X}\mathbf{g}_- - \mathbf{t} \leq \mathbf{0}, \\ & \quad \quad \quad -\mathbf{X}\mathbf{g}_+ + \mathbf{X}\mathbf{g}_- - \mathbf{t} \leq \mathbf{0}, \\ & \quad \quad \quad \mathbf{1}^\top \mathbf{t} \leq 1, \\ & \quad \quad \quad \mathbf{g}_+, \mathbf{g}_-, \mathbf{t} \geq \mathbf{0}. \end{aligned}$$

Therefore, each problem has $2p + n$ structural variables ($nvar$) and $2n + 1$ constraints ($cstr$).

We can write the problem in matrix form as

$$\begin{aligned} & \max_z \quad \mathbf{c}_i^\top \mathbf{z} \\ & \text{subject to:} \quad \mathbf{A}\mathbf{z} \leq \mathbf{b}, \quad \mathbf{z} \geq \mathbf{0}, \end{aligned}$$

where,

$$\mathbf{z} = \begin{pmatrix} \mathbf{g}_+ \\ \mathbf{g}_- \\ \mathbf{t} \end{pmatrix}, \quad \mathbf{c}_i = \begin{pmatrix} \mathbf{x}_i \\ -\mathbf{x}_i \\ \mathbf{0}_n \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} \mathbf{X} & -\mathbf{X} & -\mathbf{I}_n \\ -\mathbf{X} & \mathbf{X} & -\mathbf{I}_n \\ \mathbf{0} & \mathbf{0} & \mathbf{1}^\top \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} \mathbf{0}_n \\ \mathbf{0}_n \\ 1 \end{pmatrix}.$$

It is worth noting that it is not necessary to store $\mathbf{A}$ explicitly: the tableau can be constructed directly from $\mathbf{X}$.

To construct the simplex table and initial basis, we must first add slack variables:

$$\mathbf{A}\mathbf{z} + \mathbf{s} = \mathbf{b}, \quad \mathbf{s} \geq \mathbf{0}.$$

The total number of variables, without counting the RHS, is $nvar + cstr = 2p + 3n + 1$. The initial basis consists of the slack variables. The first n constraints have slack $\mathbf{s}^{(1)}$, the next n have $\mathbf{s}^{(2)}$, and the last one has $\mathbf{s}^{(3)}$. The initial solution

$$\mathbf{g}_+ = \mathbf{g}_- = \mathbf{t} = \mathbf{0}, \quad \mathbf{s}^{(3)} = 1$$

is feasible. For the maximization, the objective row is stored as $-\mathbf{c}_i^\top$. A variable can be included if its reduced cost is negative.

Entry and exit rules: To avoid cycles, you can use the Bland's rule, that is, go through the columns of the nonbasic variables and select the first one with a sufficiently negative reduced cost.

We must emphasize that slack variables can also be included in the basis. Therefore, the search must cover the $nvar + cstr$ variables, not just structural variables.

Problem reuse: Warm start. For each of the problems, that is, for $i = 1, \dots, n$, we have the same matrix $\mathbf{A}$ and the same vector $\mathbf{b}$. The only term that changes is

$$\mathbf{c}_i = (\mathbf{x}_i^\top, -\mathbf{x}_i^\top, \mathbf{0}^\top)^\top.$$

This is the reason why the constraints tableau is built only once. After solving the problem for one observation, the following are retained:

- the constraint tableau;
- the current basis;
- the corresponding basic feasible solution.

For the next observation, only the objective row is modified, and the simplex process continues from the current basis. This strategy avoids having to reconstruct the tableau and restart the simplex process from the slack basis for each observation.

REFERENCES

- Flores, S. (2015). Sharp non-asymptotic performance bounds for ℓ_1 and Huber robust regression estimators. *TEST* **24**, 796-812.

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